

ASSIGNMENTS

Class-9

1. Find the four fundamental subspaces of the matrix $\begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}$.
2. Find bases and the dimensions of the four fundamental spaces of the matrix

i) $\begin{bmatrix} 0 & 2 & 3 & 4 \\ 0 & 6 & 7 & 8 \\ 0 & 10 & 11 & 12 \end{bmatrix}$ ii) $\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 6 & 7 & 8 & 9 & 10 \\ 11 & 12 & 13 & 14 & 15 \end{bmatrix}$

3. Find a matrix A that has V as its row space, if V is the subspace spanned by

$$\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 5 \\ 0 \end{bmatrix}$$